DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use

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Techno-economic assessment of hydrogen export pathways from Namibia to Germany

BACKGROUND

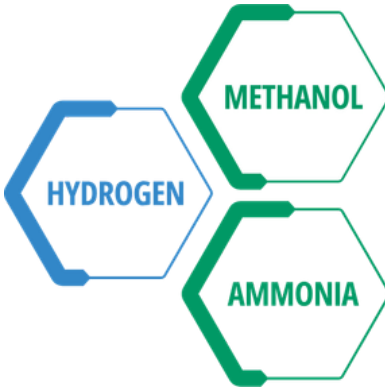
The starting point of this analysis is green hydrogen production at the Daures Green Hydrogen Village (DGHV) in Namibia. For export to international markets, several hydrogen transport and conversion options are available. To support decision-making on which export carrier should be prioritized and where the main economic levers lie, preliminary techno-economic assessments (TEA) are carried out to compare delivered cost to Germany, identify key cost drivers and sensitivities along each value chain, and indicate under which conditions (e.g., lower electricity/H_2 production costs, technology learning, market price signals) the pathways become attractive.

This combined factsheet consolidates three assessed pathways:

(i) Liquefied H_2 export with regasification at the Port of Hamburg.

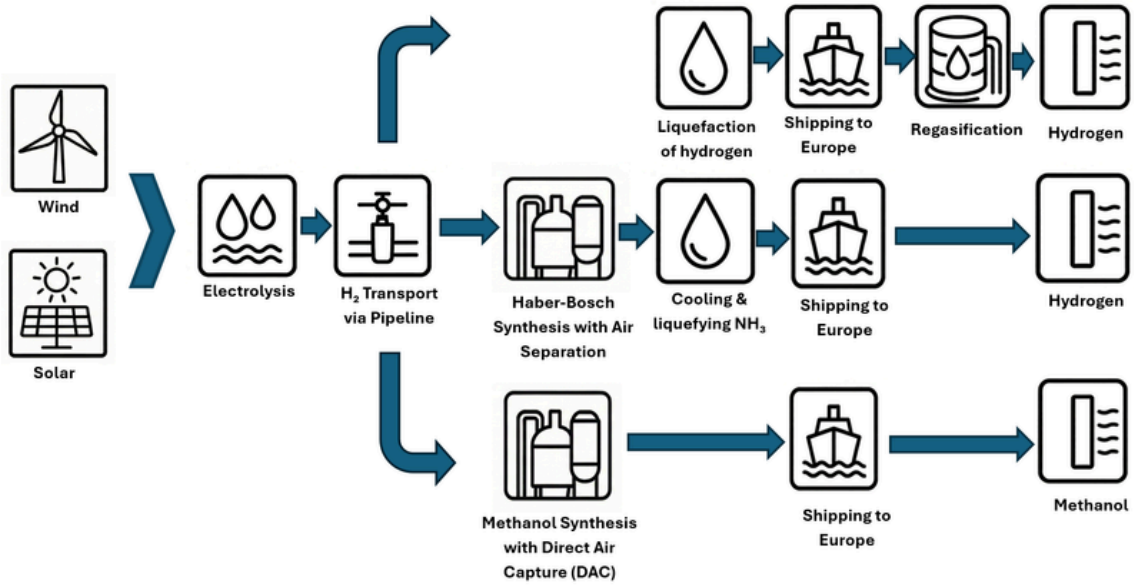
(ii) Ammonia export via Haber–Bosch synthesis with optional cracking back to hydrogen in Germany.

(iii) Methanol export produced from hydrogen and DAC-CO_2 as a liquid energy carrier and chemical feedstock.



COMPETITIVENESS VS BENCHMARKS

- Liquefied H_2: Results indicate potential cost-competitiveness with grey H_2 benchmarks, especially relevant under higher future carbon prices.
- Ammonia: Currently slightly higher than the conventional ammonia market benchmark (hence "currently uncompetitive"), but could improve with learning effects and/or higher CO_2 prices.
- Methanol: Currently, it is not economically viable, primarily due to the DAC cost dominating the chain.



ENVIRONMENTAL & STRATEGIC CONTEXT

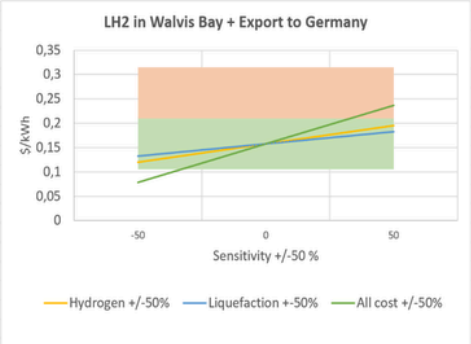
- LH_2 imports can be associated with lower GHG emissions than grey hydrogen and may justify a green premium, especially if aligned with European sustainability requirements (RED II).

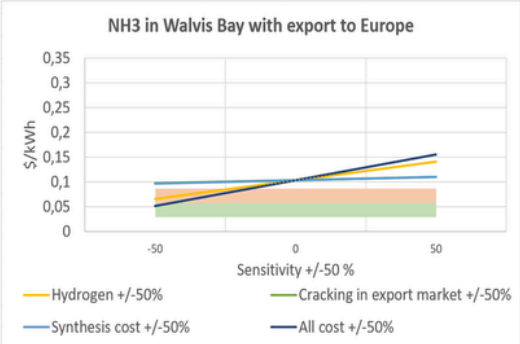
- Green ammonia can deliver lower carbon footprint benefits and may attract offtakers willing to pay a premium vs conventional grey ammonia.

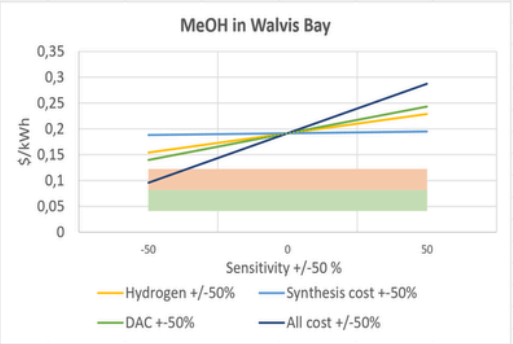
- Methanol produced using CO_2 from Direct Air Capture (DAC) can offer lower emissions than conventional methanol if the process is supplied with renewable energy, but it currently requires lower DAC costs and/or supportive market conditions to become economically viable.

HYDROGEN EXPORT PATHWAYS FROM NAMIBIA TO GERMANY: LIQUEFIED H_2 (LH_2), AMMONIA (HABER–BOSCH), AND METHANOL (DAC)

This factsheet summarizes three alternative value chains for exporting renewable hydrogen from the Daures Green Hydrogen Village (DGHV) in Namibia to Germany. First, hydrogen can be delivered as liquefied hydrogen (LH_2) by liquefaction near Walvis Bay, marine transport to Europe, and regasification at the Port of Hamburg for injection into the German gas grid. Second, hydrogen can be converted into ammonia via the Haber–Bosch process (with nitrogen from ambient air), shipped as ammonia to Europe, and optionally cracked back to hydrogen at the destination. Third, hydrogen can be combined with CO_2 captured from ambient air via Direct Air Capture (DAC) to produce methanol, which is transported as a liquid energy carrier and chemical feedstock to Europe.







MAIN COST DRIVERS (COMBINED SENSITIVITY TAKEAWAY)

Across all pathways, upstream inputs dominate total cost but the "dominant driver" differs:

- Hydrogen production cost is the biggest lever for:

1. Ammonia chain: H_2 production drives total cost; synthesis/transport/ have comparatively minor influence.

2. LH_2 chain: H_2 production has larger impact than liquefaction; liquefaction matters, but less than the cost of producing H_2.

- DAC cost dominates the methanol-with-DAC pathway:

1. DAC is the largest cost component; H_2 production is the second largest; reducing either greatly improves competitiveness.

KEY RESULTS

Delivered cost indicators (preliminary):

Pathway	Delivered product	Cost result stated in factsheets
Ammonia (Haber-Bosch)	Gaseous H_2	**~10 ct/kWh (2024)** gaseous H_2 in Germany
Liquefied H_2	Gaseous H_2	**~ 15.7 ct/kWh (2024)** delivered H_2 in Germany
Methanol (Direct Air Capture)	Liquid methanol	**~ 20 ct/kWh (2024)** MeOH in Germany

CONCLUSIONS

Across the three export pathways, the preliminary TEA results highlight that upstream production costs (cost are alculated using hydrogen-export tool by the Institute for Energy Economics EWI, Cologne [1]) are the main lever for competitiveness, but the dominant driver differs by pathway: hydrogen production cost is most influential for both LH_2 export and the ammonia route (including cracking), while DAC cost dominates the methanol route (with hydrogen cost as the second-largest contributor).

For the **ammonia pathway** (including cracking back to hydrogen in Germany), the overall cost along the value chain equates to roughly 10 ct/kWh of gaseous hydrogen in Germany (preliminary result). The sensitivity analysis (±50% variation of cost components) shows that the initial hydrogen production has the biggest influence on the overall cost, while synthesis, transport, and cracking have comparatively minimal effects. Benchmark stripes in the results illustrate the world market ammonia cost and its ±50% price sensitivity; the preliminary outcome indicates that exporting ammonia from Namibia to Germany is currently slightly higher in cost than conventional ammonia and therefore currently uncompetitive with market prices under the assumptions used.

For the **LH_2 pathway**, findings show that the overall costs along the liquefied hydrogen value chain equate to 15.7 (2024) ct/kWh [2] for gaseous hydrogen delivered to the national gas grid of Germany. The sensitivity analysis indicates that the production costs for hydrogen via electrolysis have a higher impact on total delivered cost than the liquefaction process. In comparison with a benchmark for grey hydrogen produced in Germany (steam methane reforming), the results indicate that exporting hydrogen from Namibia to Germany as LH_2 can be competitive, particularly under higher future carbon prices.

For the **methanol pathway**, the overall cost along the value chain equates to roughly 20 ct/kWh (2024) of liquid methanol in Germany (preliminary result), calculated with the same EWI hydrogen-export-tool. The sensitivity analysis (±50% variation) confirms DAC as the dominant cost driver, with hydrogen production itself being the second largest cost component. The benchmark stripes represent the current market price of methanol and its ±50% price sensitivity; under present assumptions the pathway is not economically viable, and competitiveness improves strongly if DAC and hydrogen production costs decline.

REFERENCES

[1]EWI May 2024; available online:https://www.ewi.uni koeln.de/en/publications/ewi-global-ptx-cost-tool/

[2] Hank, C., Holst, M., Thelen, C., Kost, C., Längle, S., Schaadt, A., & Smolinka, T. (2023). Site-specific, comparative analysis for suitable Power-to-X pathways and products in developing and emerging countries. Fraunhofer Institute for Solar Energy Systems ISE.

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https://github.com/IER-Hy4Daures/Fact-Sheets